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ABSTRACT

The exam preparation processes for engineering and quantitative courses in educational institutions consume teachers' time and create a significant workload. In traditional exam methods, administering the exact same questions to all students compromises exam security, increases the risk of cheating, and brings the fairness of the evaluations into question.

This project aims to increase trust in exam results, minimize the time lost while creating unique exam papers, and enhance fairness by providing solutions to these core problems. In line with these objectives, UniTest, a web-based automatic exam variation platform, has been developed. It has become a project that leaves tedious manual processes behind.

UniTest aims to generate different exam questions of the same difficulty level without altering the logic of the question, by using artificial intelligence to modify the numerical data within a single main question stem, question text, exam paper, or question image. This application is built upon a Flask-based web backend, an SQLite database, the GPT-4o-mini large language model, and a PDF generation engine integrated with the ReportLab library. Thanks to its multimodal artificial intelligence infrastructure, the project can automatically select, extract, and modify numerical values from exam paper images. Teachers can generate bulk exam papers based on a specified number of students, automatically produce answer keys by entering a reference Teacher's Solution into the system, and display as well as manage past exams through a question bank.

In conducted tests, UniTest successfully modified numerical questions, particularly in Physics courses, drew electrical circuits, and produced unique variations of exam papers. By automating teachers' exam preparation processes, this project provides significant time efficiency, increases the reliability of examinations, and offers a concrete example of the practical use of large language models in the field of education.

Keywords: Artificial Intelligence, Reliability, Fairness, Quantitative, Variation